LOG 6309E - Intelligent DevOps

Group Assignment: *Log-based Anomaly Detection*

Notes

- This assignment is done in a group of three students.
- Weight: 15% of the final grade.
- Deliverable: An IEEE report of 6-8 pages in PDF. A template is available on Moodle.
- Assignment Presentation – Friday, OCT 24 (submit slides by OCT 23 23:59pm)
- Assignment Report Due – Monday, NOV 3 11:59 PM

Objectives

- Understand and apply the concepts and techniques of log parsing.
- Understand and apply the concepts and techniques of log-based anomaly detection.
- Perform an empirical evaluation study of existing techniques.
- Understand and apply basic data analysis techniques.
- Improve the skill of writing a scientific report.
- Practice teamwork in a research project.

Specification

This assignment requires you to partially replicate and extend a recent paper [1] related to log-based anomy detection. The paper describes a pipeline for log-based anomaly detection and evaluates the impact of different components (parsing, representation, etc.) within the pipeline.

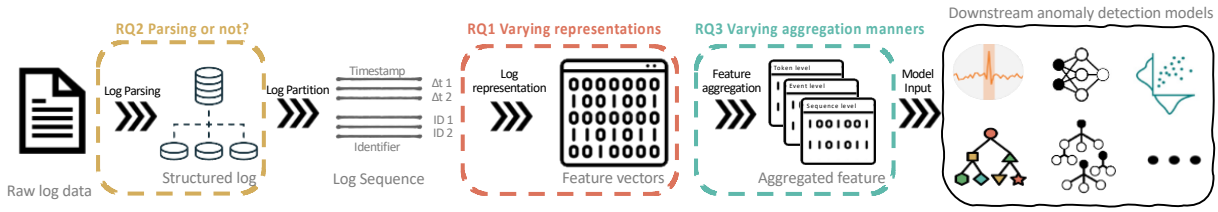


Figure 1: A pipeline for log-based anomaly detection [1]

[50% points] Replication. Replicate RQ1 of the paper [1] – build log-based anomaly detection models through the following three steps.

1) [15%] Parse the raw log data into event templates (*a.k.a.*, log parsing or log template extraction). You can use Drain or other log parsers (you may find a list of log parsers here). Note: pre-processing and post-processing may be needed before/after using a log parser.

    a. You can consider the pre-processing approach used in Qin et al.[1].
2) [15%] Represent the parsed log data as numeric features. Choose the appropriate log representation methods (e.g., *Message Count Vector*) according to your downstream models. You only need to consider one representation method for each model.
3) [20%] Construct and evaluate log-based anomaly detection models. You are required to consider two supervised models, including one classical model and one deep leaning model. You should use four standard evaluation metrics: precision, recall, F1-score and AUC).
    a. Consider applying correlation analysis (e.g., hierarchical clustering) and redundancy analysis (e.g., variance inflation factor, or VIF analysis) to remove collinearity and multi-collinearity among the features before constructing the models. Compare the performance of the models with and without this step.

You should perform the experiments on two log datasets, out of the four used in the paper. You can leverage the replication package of the paper in your work. Compare your results with that in the original paper.

[50% points] Extension. Extend the analyses performed in the replication. Perform the following analyses on one log dataset.

1) [20%] Construct and evaluate two unsupervised/semi-supervised models/approaches; cover two categories from classical models, deep learning models, and LLM-based approaches. Check the "Anomaly detection models to be considered" section[2] for candidate models/approaches.
2) [15%] Use statistical ranking (e.g., Scott-Knott test) to rank the four models/approaches by their performance (e.g., F1-score or AUC). Note: you can produce a statistical distribution of the performance of each model/approach through resampling the training/testing datasets multiple times (e.g., 10 times).
3) [15%] Focusing on the best performing model/approach, analyze and explain the model (e.g., important features, SHAP values, LIME, or LLM explanation techniques[3]). You may consider a different model/approach but explain your rationale (e.g., the one with the best interpretability).

Resources

Paper to be replicated and extended:

[1] X. Wu, H. Li, and F. Khomh, "On the Effectiveness of Log Representation for Log-based Anomaly Detection," *Empirical Software Engineering* (EMSE), 2023. PDF

[1] Qin, Qiaolin, et al. "Preprocessing is All You Need: Boosting the Performance of Log Parsers With a General Preprocessing Framework." 2025 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER). IEEE, 2025.
[2] You may also consider other approaches not in the list.
[3] Zhao, Haiyan, et al. "Explainability for large language models: A survey." ACM Transactions on Intelligent Systems and Technology 15.2 (2024): 1-38.

Anomaly detection models to be considered:

Classical models:

Shilin He, Jieming Zhu, Pinjia He, and Michael R. Lyu. "**Experience report: System log analysis for anomaly detection**." In 2016 IEEE 27th International Symposium on Software Reliability Engineering (ISSRE), pp. 207-218. IEEE, 2016.

Repo: https://github.com/logpai/loglizer

Yu, Boxi, et al. "**Deep learning or classical machine learning? An empirical study on log-based anomaly detection**." Proceedings of the 46th IEEE/ACM international conference on software engineering. 2024.

Deep learning models:

Chen, Zhuangbin, Jinyang Liu, Wenwei Gu, Yuxin Su, and Michael R. Lyu. "**Experience Report: Deep Learning-based System Log Analysis for Anomaly Detection**." arXiv preprint arXiv:2107.05908 (2021).

Repo: https://github.com/logpai/deep-loglizer

LLM-based approaches:

Zhang, Wanhao, et al. "Leveraging RAG-Enhanced Large Language Model for Semi-Supervised Log Anomaly Detection." 2024 IEEE 35th International Symposium on Software Reliability Engineering (ISSRE). IEEE, 2024.

Repo: https://github.com/WanhaoZhang/LogRAG

Han, Xiao, Shuhan Yuan, and Mohamed Trabelsi. "Loggpt: Log anomaly detection via gpt." 2023 IEEE International Conference on Big Data (BigData). IEEE, 2023.

Ma, Lipeng, et al. "AdaptiveLog: An Adaptive Log Analysis Framework with the Collaboration of Large and Small Language Model." ACM Transactions on Software Engineering and Methodology (2025).

Datasets:

Use two of the datasets used in the replicated paper [1]. The datasets can be found in the following repos:

https://github.com/logpai/loghub

https://www.usenix.org/cfdr-data

Writing tips

- Use the "Assignment report template" available on Moodle.
- Follow the writing tips discussed in the course.

- Describe your approach in a concise and unambiguous way: others should be able to repeat your experiments following your report.
- Justify your design decisions (the selection of the models, representation techniques, evaluation metrics, and statistical analysis methods).
- Don’t just present your results, but also explain your results and discuss their implications. If you cannot explain a result, then something is probably wrong.
- Highlight a few take-home messages that you want readers to learn from your results.